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Assignment-2

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1. Given vectors $a = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, find the projection of b onto a, the perpendicular component, and verify that

$$b = b_{\parallel a} + b_{\perp a}.$$

Ans:

Here, $a = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Projection of b onto a:-

$$b_{\parallel a} = \frac{b^T a}{a^T a} a$$

$$b^T a = \begin{bmatrix} 3 \\ 4 \end{bmatrix}^T \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= 3 \times 2 + 4 \times 1$$

$$= 6 + 4$$

$$= 10$$

$$a^T a = \begin{bmatrix} 2 \\ 1 \end{bmatrix}^T \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= 2 \times 2 + 1 \times 1$$

$$= 4 + 1$$

$$= 5$$

Now, $b_{\parallel a} = \frac{b^T a}{a^T a} a = \frac{10}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$= 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

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Perpendicular component:-

$$b_{\perp a} = b - b_{\parallel a}$$

$$= \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3-4 \\ 4-2 \end{bmatrix}$$

$$= \begin{pmatrix} -1 \\ +2 \end{pmatrix}$$

Verification:-

$$b = b_{\parallel a} + b_{\perp a}$$

or, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

or, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4-1 \\ 2+2 \end{bmatrix}$

or, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (True)

$\therefore$ $b = b_{\parallel a} + b_{\perp a}$, verified.

2. For $a = \begin{bmatrix}1\\2\\2\end{bmatrix}$, $b = \begin{bmatrix}2\\0\\1\end{bmatrix}$, compute the projection of b onto a, the angle between a and b, and verify orthogonality of the residual vector.

Ans:

Here, $a = \begin{bmatrix}1\\2\\2\end{bmatrix}$ and $b = \begin{bmatrix}2\\0\\1\end{bmatrix}$

<u>Projection of b onto a:-</u>

$$b_{\|a} = \frac{b^T a}{a^T a} a$$

Now, $\cancel{b_{\|a} = \frac{\begin{bmatrix}2\\0\\1\end{bmatrix}^T \begin{bmatrix}1\\2\\2\end{bmatrix}}{\quad}}$

$$b_{\|a} = \frac{\begin{bmatrix}2\\0\\1\end{bmatrix}^T \begin{bmatrix}1\\2\\2\end{bmatrix}}{\begin{bmatrix}1\\2\\2\end{bmatrix}^T \begin{bmatrix}1\\2\\2\end{bmatrix}} \begin{bmatrix}1\\2\\2\end{bmatrix}$$

$$= \frac{\begin{bmatrix}2 & 0 & 1\end{bmatrix} \begin{bmatrix}1\\2\\2\end{bmatrix}}{\begin{bmatrix}1 & 2 & 2\end{bmatrix} \begin{bmatrix}1\\2\\2\end{bmatrix}} \begin{bmatrix}1\\2\\2\end{bmatrix}$$

$$= \frac{2\times1 + 0\times2 + 1\times2}{1\times1 + 2\times2 + 2\times2} \begin{bmatrix}1\\2\\2\end{bmatrix}$$

$$= \frac{4}{9} \begin{bmatrix}1\\2\\2\end{bmatrix}$$

$$\therefore b_{\|a} = \begin{bmatrix}4/9\\8/9\\8/9\end{bmatrix}$$ #

Angle between a and b:-

If θ be the angle between 'a' and 'b' then

$$\cos\theta = \frac{a^T b}{\|a\|\,\|b\|}$$

Here,

$$\|a\| = \sqrt{1^2+2^2+2^2}$$
$$= \sqrt{1+4+4}$$
$$= \sqrt{9}$$
$$= 3$$

$$\|b\| = \sqrt{2^2+0^2+1^2}$$
$$= \sqrt{4+0+1}$$
$$= \sqrt{5}$$

Now,

$$\cos\theta = \frac{\begin{bmatrix}1\\2\\2\end{bmatrix}^T \begin{bmatrix}2\\0\\1\end{bmatrix}}{\|a\|\,\|b\|}$$

$$= \frac{\begin{bmatrix}1 & 2 & 2\end{bmatrix}\begin{bmatrix}2\\0\\1\end{bmatrix}}{3\times\sqrt{5}}$$

$$= \frac{1\times2+2\times0+2\times1}{3\sqrt{5}}$$

$$= \frac{2+2}{3\sqrt{5}}$$

$$= \frac{4}{3\sqrt{5}}$$

or, $\theta = \cos^{-1}\left(\frac{4}{3\sqrt{5}}\right)$

$\therefore \theta = 53.3^\circ$ #

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Angle between a and b :-

If θ be the angle between 'a' and 'b' then

$$\cos\theta = \frac{a^T b}{\|a\| \|b\|}$$

Here,

$$\|a\| = \sqrt{1^2+2^2+2^2}$$
$$= \sqrt{1+4+4}$$
$$= \sqrt{9}$$
$$= 3$$

$$\|b\| = \sqrt{2^2+0^2+1^2}$$
$$= \sqrt{4+0+1}$$
$$= \sqrt{5}$$

Now,

$$\cos\theta = \frac{\begin{bmatrix}1\\2\\2\end{bmatrix}^T \begin{bmatrix}2\\0\\1\end{bmatrix}}{\|a\| \|b\|}$$

$$= \frac{\begin{bmatrix}1 & 2 & 2\end{bmatrix}\begin{bmatrix}2\\0\\1\end{bmatrix}}{3 \times \sqrt{5}}$$

$$= \frac{1\times2+2\times0+2\times1}{3\sqrt{5}}$$

$$= \frac{2+2}{3\sqrt{5}}$$

$$= \frac{4}{3\sqrt{5}}$$

or, $\theta = \cos^{-1}\left(\frac{4}{3\sqrt{5}}\right)$

$\therefore \theta = 53.3^\circ$ #

Residual :-

If residual is 'r' then

$$r = b - b_{\|a}$$

$$= \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 4/9 \\ 8/9 \\ 8/9 \end{bmatrix}$$

$$= \begin{bmatrix} 2-4/9 \\ 0-8/9 \\ 1-8/9 \end{bmatrix}$$

$$\therefore r = \begin{bmatrix} 14/9 \\ -8/9 \\ 1/9 \end{bmatrix} \quad \#$$

Verification of orthogonality :-

For orthogonality,

$$a^T r = 0$$

Now,

$$a^T r = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}^T \begin{bmatrix} 14/9 \\ -8/9 \\ 1/9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 14/9 \\ -8/9 \\ 1/9 \end{bmatrix}$$

$$= 1 \times \frac{14}{9} + 2 \times \left(\frac{-8}{9}\right) + 2 \times \frac{1}{9}$$

$$= \frac{14-16+2}{9}$$

$$= 0$$

So, residual is perpendicular to a.

3. Show that the projection matrix onto a vector 'a' is given by

$$P = \frac{aa^T}{a^T a}$$ and prove that $P^2 = P$.

Ans:

We know,

Projection of b onto a:

$$\text{Proj}_a(b) = \frac{a^T b}{a^T a} a$$

$$(\because a^T b = b^T a)$$

Rewriting above formula:-

Projection of b onto a: $\text{proj}_a(b) = \frac{aa^T}{a^T a} b$

$$\therefore P = \frac{aa^T}{a^T a}$$

Now, $$P^2 = \left(\frac{aa^T}{a^T a}\right)\left(\frac{aa^T}{a^T a}\right)$$

or, $$P^2 = \frac{a(a^T a)a^T}{(a^T a)^2}$$

or, $$P^2 = \frac{aa^T}{a^T a}$$

or, $P^2 = P$, proved #

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4. Show that projection minimizes the distance $\|b-\alpha a\|^2$ and find the optimal value of α.

Ans:-

Let $f(\alpha) = \|b-\alpha a\|$

$$= (b-\alpha a)^T (b-\alpha a)$$

$$= (b^T - \alpha a^T)(b-\alpha a) \quad (\because (a-b)^T = a^T - b^T)$$

$$= b^T b - \alpha b^T a - \alpha a^T b + \alpha^2 a^T a$$

Since, $b^T a = a^T b$.

$$\therefore f(\alpha) = b^T b - 2\alpha a^T b + \alpha^2 a^T a$$

Now, differentiating with respect to α; we have

$$\frac{df}{d\alpha} = -2a^T b + 2\alpha a^T a$$

For minimum distance: $\frac{df}{d\alpha} = 0$

$$\therefore -2a^T b + 2\alpha a^T a = 0$$

or, $\boxed{\alpha = \frac{a^T b}{a^T a}}$

Again, take the second derivative: $\frac{d^2 f}{d\alpha^2}$

$$\frac{d^2 f}{d\alpha^2} = 2a^T a$$

As, $a^T a = \|a\|^2 > 0$

$$\therefore \frac{d^2 f}{d\alpha^2} > 0$$

Hence, the function is minimized.

5. Find eigenvalues of $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and interpret them geometrically.

Ans:

Eigenvalues of A:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Now, $\det(A - \lambda I) = 0$, λ = eigenvalue

or, $\left| \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$

or, $\begin{vmatrix} 2-\lambda & 1-0 \\ 1-0 & 2-\lambda \end{vmatrix} = 0$

or, $(2-\lambda)^2 - 1^2 = 0$

or, $(2-\lambda)^2 = 1^2$

or, $2-\lambda = \pm 1$

$\therefore \lambda = 1, 3$

Geometrical Interpretation:-

Eigenvectors are directions that do not rotate. They only stretch or shrink. In above calculation, for $\lambda = 3$, one direction is stretched by 3 factor and for $\lambda = 1$, and perpendicular direction is stretched by 1.

6. For $A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ compute eigenvalues and explain how $x^T A x$ scales space.

Ans:

Here, $A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

For eigenvalues, $\det(A - \lambda I) = 0$

or, $\left| \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$

or, $\begin{vmatrix} 3-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0$

or, $(3-\lambda)(1-\lambda) - 0 = 0$

Either, $3-\lambda = 0 \Rightarrow \lambda = 3$

OR, $1-\lambda = 0 \Rightarrow \lambda = 1$

$\therefore \boxed{\lambda = 1, 3}$

For $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Now, $x^T A x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$= \begin{bmatrix} 3x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$= 3x_1^2 + x_2^2$

The result of $x^T A x$ is $3x_1^2 + x_2^2$

and this indicates than x-direction is stretched more strongly than y-direction.

7. Explain why eigenvectors represent directions in data.

Ans:

In PCA, we use the covariance matrix. The covariance matrix tells how data spreads in different directions. Eigenvectors of the covariance matrix give the directions of spread. Eigenvalues tell how much spread exists in those directions. The eigenvector with the largest eigenvalue is the first principal direction because data varies the most along that direction.

8. Define Bernoulli distribution and derive its mean and variance.

Ans:-

A Bernoulli random variable has only two outcomes:

$$X = \begin{cases} 1, & \text{success} \\ 0, & \text{failure} \end{cases}$$

$$P(X=1) = 1$$

$$P(X=0) = 1-P$$

Now,

$$\begin{aligned} \text{Mean} &= E(X) \\ &= 1 \times p + 0 \times (1-P) \\ &= p + 0 - P \\ &= P \end{aligned}$$

Variance:

$$Var(X) = E[X^2] - (E[X])^2$$

Since

$$X^2 = X,$$

$$E[X^2] = p$$

$$Var(X) = p - p^2$$

$$= p(1-p)$$

9. For $X \sim$ Bernoulli (p), compute mean and variance when $p = 0.3$, and interpret the result in a classification setting.

Ans:

Here, $X \sim$ Bernoulli (p)

$$p = 0.3$$

$$E(X) = p = 0.3$$

$$Var(X) = p(1-p)$$

$$= 0.3(1-0.3)$$

$$= 0.3 \times 0.7$$

$$= 0.21$$

Interpretation:-

In classification, if success means, class = 1, then the model gives class 1 with probability 0.3. The variance 0.21 shows uncertainty in this binary outcome.

10. Show that Bernoulli distribution is a special case of Binomial distribution.

Ans:

If $n=1$, then Binomial distribution becomes Bernoulli distribution.

Here, $n \to$ no. of trials.

$$X \sim \text{Binomial}(1, p)$$

$$P(X=k) = \binom{1}{k} p^k (1-p)^{1-k}$$

if $k=0$

$$P(X=0) = \binom{1}{0} p^0 (1-p)^{1-0}$$

$$= \binom{1}{0} (1-p) = 1(1-p)$$

$$= (1-p)$$

if $k=1$

$$P(X=1) = \binom{1}{1} p^1 (1-p)^{1-1}$$

$$= \binom{1}{1} p (1-p)^0$$

$$= \binom{1}{1} p = 1 p = p$$

This is exactly Bernoulli.

11. Derive the Binomial distribution formula

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

from Bernoulli trials.

Ans:

For n independent Bernoulli trials, suppose exactly k successes occur.

Probability of one specific arrangement: $p^k (1-p)^{n-k}$

Number of ways to choose which k trials are successes:

$$\binom{n}{k}$$

$$\therefore \quad P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \#$$

12. For $n=5$, $p=0.4$, compute $P(X=0)$, $P(X=2)$, and $P(X\geq 3)$.

Ans:

Here, $n=5$, $p=0.4$ and $q=0.6$

Now,

$$P(X=0) = \binom{5}{0} (0.4)^0 (0.6)^{5-0}$$

$$= \binom{5}{0} \times 1 \times (0.6)^5$$

$$= \frac{5!}{0!\,(5-0)!} \times (0.6)^5$$

$$= 1 \times (0.6)^5$$

$$= 0.07776$$

Again,

$$P(X=2) = \binom{5}{2}(0.4)^2(0.6)^{5-2}$$

$$= \frac{5!}{2!(5-2)!}(0.4)^2(0.6)^3$$

$$= \frac{5\times4\times3!}{2\times3!}\times0.16\times0.216$$

$$= \frac{20}{2}\times0.16\times0.216$$

$$= 10\times0.16\times0.216$$

$$= 0.3456$$

$$P(\geq 3) = P(3) + P(4) + P(5)$$

$$= \binom{5}{3}(0.4)^3(0.6)^{5-3} + \binom{5}{4}(0.4)^4(0.6)^{5-4} + \binom{5}{5}(0.4)^5(0.6)^{5-5}$$

$$= \frac{5!}{3!2!}\times(0.4)^3(0.6)^2 + \frac{5!}{4!1!}(0.4)^4(0.6)^1 + \frac{5!}{5!0!}(0.4)^5(0.6)^0$$

$$= \frac{5\times4\times3!}{3!2!}\times0.064\times0.36 + \frac{5\times4!}{4!}(0.0256)(0.6) + 1\times(0.4)^5\times1$$

$$= 10\times0.064\times0.36 + 5\times0.0256\times0.6 + 0.01024$$

$$= 0.2304 + 0.0768 + 0.01024$$

$$\therefore P(X\geq3) = 0.31744 \quad \text{Ans}$$

13. Show that $E[X] = np$

$Var(X) = np(1-p)$ for a Binomial distribution

Ans:

Let $X = X_1 + X_2 + \cdots\cdots + X_n$

where

$X_i \sim$ Bernoulli (p).

$\therefore$

$E[X_i] = p,$

Also, $Var(X_i) = p(1-p)$

Now,

Mean:

$$E(X) = E[X_1 + X_2 + X_3 + \cdots\cdots X_n]$$
$$= E(X_1) + E(X_2) + E(X_3) + \cdots E(X_n)$$
$$= p + p + p + \cdots p$$
$$= np$$

$$\therefore E(X) = np$$

Again, since for variance, every trial is independent

$$\therefore Var(X) = Var(X_1) + Var(X_2) + \cdots + Var(X_n)$$
$$= p(1-p) + p(1-p) + \cdots + p(1-p)$$
$$= np(1-p)$$

$$\therefore Var(X) = np(1-p)$$

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14. Write the probability density function of the normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

and explain each parameter.

Ans:

The probability density function of the normal distribution function is given by formula:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$$

Explanation of the parameters:-

μ :- The mean, it is the arithmetic average of the distribution.

σ : The standard deviation, it measures the dispersion or spread of the data.

x : The variable, it is a specific data point.

π : Pi, a mathematical constant.

e : Euler's number, a constant

15. For $\mu = 0$, $\sigma^2 = 1$, write the standard normal distribution and compute the z-score transformation

$$Z = \frac{x-\mu}{\sigma}$$

Ans: Here,

$\mu = 0$

$\sigma^2 = 1$

$\therefore \sigma = \sqrt{\sigma^2}$

$= \sqrt{1}$

$= 1$

So, $Z \sim N(0,1)$

Pdf for standard normal distribution $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

For Z-score transformation:

$$Z = \frac{X-\mu}{\sigma}$$

if $\mu = 0$ and $\sigma = 1$ then

$$Z = \frac{X-0}{1}$$

$$\Rightarrow \boxed{Z = X}$$

16. Explain the empirical rule (68-95-99.7) for normal distribution.

Ans:

Here, the empirical rule (68-95-99.7) explains that 68% of data lies within $\mu \pm 1\sigma$. In addition, 95% data lies within $\mu \pm 2\sigma$ and 99.7% of data lies within $\mu \pm 3\sigma$. So, basically this empirical rule explains about the data spreadness around the mean and the probability of data to be around the mean.

17. Explain why normal distribution appears in measurement errors, PCA, and the central Limit Theorem.

Ans:

Normal distribution appears in measurement errors because many small independent errors add together. It appears in the CLT because the sum or average of many independent random variables tends toward normality.

18. Explain how covariance matrix leads to eigenvalue decomposition in PCA.

Ans: In PCA, covariance matrix represents the variance and correlation among features.

$$C = \frac{1}{n-1} X^T X$$

X = centered data matrix

n = no. of samples.

PCA finds the directions of maximum variance by solving the eigenvalue equation:

$Cu = \lambda u$; u = eigenvector

λ = eigenvalue

19. Explain the relationship between vector projection, maximum variance direction, and the largest eigenvalue of the covariance matrix.

Ans: In PCA, the main objective is to find the direction in which the data has maximum variance. This direction is obtained using the eigenvectors and eigenvalues of the covariance matrix.

In PCA, data is projected onto a direction u.

The projection of a data vector x is:

$$z = u^T x$$

The variance of the projected data is:

$$Var(z) = u^T C u$$

C = covariance matrix

PCA find the direction u that maximizes the projected vector.

Solving this optimization gives the eigenvalue equation:

$$C\lambda = \lambda u.$$

The eigenvalue represents the variance along its corresponding eigenvector:

$$u^T C u = \lambda$$

Therefore, the eigenvector with the ~~target~~ largest eigenvalue gives the direction of maximum value and is chosen as the first principal component.

20. Given $x_1 = [1, 2]$, $x_2 = [2, 3]$, $x_3 = [3, 5]$, compute the mean vector, center the data, find the covariance matrix, compute eigenvalues, and interpret the first principal direction.

Soln:

Here, $x_1 = [1, 2]$
$x_2 = [2, 3]$
$x_3 = [3, 5]$

(a) $$\text{mean}(\mu) = \left[\frac{1+2+3}{3}, \frac{2+3+5}{3}\right]$$
$$= [2, 10/3]$$
$$\therefore \mu = \begin{bmatrix} 2 \\ 10/3 \end{bmatrix}$$

Now,

(b) centered data (X_c) $$= \begin{bmatrix} 1-2 & 2-10/3 \\ 2-2 & 3-10/3 \\ 3-2 & 5-10/3 \end{bmatrix}$$
$$= \begin{bmatrix} -1 & -4/3 \\ 0 & -1/3 \\ 1 & 5/3 \end{bmatrix}$$

(c) Calculating co-variance matrix:-

$$\text{Cov}(C) = \frac{1}{n-1} X_c^T X_c$$
$$= \frac{1}{3-1} \begin{bmatrix} -1 & -4/3 \\ 0 & -1/3 \\ 1 & 5/3 \end{bmatrix}^T \begin{bmatrix} -1 & -4/3 \\ 0 & -1/3 \\ 1 & 5/3 \end{bmatrix}$$
$$= \frac{1}{2} \begin{bmatrix} -1 & 0 & 1 \\ -4/3 & -1/3 & 5/3 \end{bmatrix} \begin{bmatrix} -1 & -4/3 \\ 0 & -1/3 \\ 1 & 5/3 \end{bmatrix}$$
$$= \begin{bmatrix} (1+1)/2 & (4/3+0+5/3)/2 \\ (4/3+5/3)/2 & (16/9+\frac{1}{9}+\frac{25}{9})/2 \end{bmatrix}$$

or, $Cov(C) = \begin{bmatrix} 1 & 3/2 \\ 3/2 & 7/3 \end{bmatrix}$

(d) Now, finding eigenvalues

$$\det(C - \lambda I) = 0$$

$$\text{or, } \begin{vmatrix} 1-\lambda & 3/2 \\ 3/2 & 7/3-\lambda \end{vmatrix} = 0$$

$$\text{or, } (1-\lambda)(7/3-\lambda) - 9/4 = 0$$

$$\text{or, } 7/3 - \frac{7\lambda}{3} - \lambda + \lambda^2 - 9/4 = 0$$

$$\text{or, } \frac{7 - 7\lambda - 3\lambda + 3\lambda^2}{3} = 9/4$$

$$\text{or, } 3\lambda^2 - 10\lambda + 7 = 27/4$$

$$\text{or, } 12\lambda^2 - 40\lambda + 28 = 27$$

$$\text{or, } 12\lambda^2 - 40\lambda + 28 - 27 = 0$$

$$\text{or, } 12\lambda^2 - 40\lambda + 1 = 0$$

Above equation is in the form of $ax^2 + bx + c = 0$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{So, } \lambda = \frac{-(-40) \pm \sqrt{(-40)^2 - 4.12.1}}{2.12}$$

or, $\lambda = \dfrac{40 \pm \sqrt{1600 - 48}}{24}$

or, $\lambda = \dfrac{40}{24} \pm \dfrac{\sqrt{1552}}{24}$

Taking positive sign;

$$\lambda = \frac{40}{24} + \frac{\sqrt{1552}}{24}$$
$$= 3.308$$

Taking negative sign;

$$\lambda = \frac{40}{24} - \frac{\sqrt{1552}}{24}$$
$$= 0.025$$

(e) Here, $\lambda_1 = 3.308$
$\lambda_2 = 0.025$

The first eigenvalue is much larger and the second one ~~ver~~ very small. So, most variation lies along the first principal direction.

Finding eigenvector for $\lambda_1 = 3.308$:-

$$C\vec{v} = \lambda\vec{v} \quad (\because C = \text{covariance matrix})$$

or, $(C - \lambda I)\vec{v} = \vec{0}$ $\quad \left[\because \vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right]$

or, $\left(\begin{pmatrix} 1 & 3/2 \\ 3/2 & 7/3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right)\vec{v} = \vec{0}$

or, $\begin{pmatrix} 1-\lambda & 3/2 \\ 3/2 & 7/3-\lambda \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

or, $\begin{pmatrix} 1-3.308 & 1.5 \\ 1.5 & 2.33-3.308 \end{pmatrix}\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

or, $\begin{pmatrix} -2.308 & 1.5 \\ 1.5 & -0.975 \end{pmatrix}\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

or $\begin{pmatrix} -2.308V_1 + 1.5V_2 \\ 1.5V_1 \cancel{+} -0.975V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

From above, we have

$$-2.308V_1 + 1.5V_2 = 0 \quad \text{——— (i)}$$
$$1.5V_1 - 0.975V_2 = 0 \quad \text{——— (ii)}$$

Solving (i) and (ii), we have

$V_1 \approx 0.650$ (After normalizing)

$V_2 \approx 0.760$ (After normalizing)

Interpretation:

The data mainly spreads in an upward diagonal direction. Since the first eigenvalues is much larger than the second, the 2 dimensional data can be well represented by projecting it onto first principal direction.